Learning Interpretable Pit-Stop Policies for Formula 1 from Observational Race Data

15.095 Project Proposal — Georgia Abou Chebel, Nanrui Gong — October 27, 2026

1. Motivation

Pit-stop timing and tyre choice are among the most consequential decisions in a Formula 1 race. A stop costs about twenty seconds of lap time at most circuits, but fresh tyres can win much of it back, and the timing decides track position through the undercut and overcut. Teams decide using private simulators, driver feedback and live telemetry. Public work mostly optimises strategy *inside a simulator*: discrete-event simulation [1], Monte Carlo race simulation and a neural "virtual strategy engineer" [4, 5], and a mixed-integer program benchmarking a reinforcement-learning agent. Other work *predicts* when teams will pit from telemetry. Prediction is not prescription: the data reflect what teams did, not what was best.

Question. Given only what a strategist sees at a decision point (tyre state, race phase, gaps, track status, weather), can we learn a decision rule for "stay out" versus "pit" that is *estimated* to beat historical team behaviour on races reserved for test?

Plan. We treat historical stops as logged decisions in an observational study. We estimate counterfactual outcomes, correct for non-random team behaviour, and fit an optimal policy tree of depth 2–3, yielding rules like "if tyre age > 25 and gap behind < 2 s, pit."

Data. We use the free OpenF1 REST API [6]. Its `laps`, `stints`, `pit`, `intervals`, `position`, `race_control` and `weather` endpoints cover lap times, tyre compound and age at stint start, pit events, gaps, running order, track status and weather; its sessions and results endpoints allow race-level splits. From the documentation we expect everything we need except the gap to the car *behind*, which we derive from the interval data of the following car.

2. Proposed Methods

Reduction to a one-shot decision. A race is sequential but a policy tree picks one action per row of a reward matrix Γ. We create one row per (driver, lap) *decision point* at which a stop is a real option (e.g. tyre age above a minimum, not already in the pit lane). Context x_i: compound, tyre age, compounds used so far, fraction of race, gaps ahead and behind, position, track status, weather, circuit. Actions: stay out or pit (compound choice added in a second stage). Outcome y_i: driver's time advantage over the field median across the next K laps, so track-wide events largely cancel. This is *myopic*: a fixed horizon ignores how today's choice shapes later stops. We state this as a limitation, not a claim of full race optimality.

Pipeline. Let T_i be the action taken.

1. *Outcome models.* Fit $f_t(x) \approx \mathbb{E}[y \mid x, T = t]$ with regularised regression or optimal regression trees, cross-fitted by race [3].
2. *Propensities.* Teams are not randomised and probabilities are unknown, so estimate $p_{it} = \Pr(T_i = t \mid x_i)$ with logistic regression or a calibrated classifier.
3. *Rewards.* $\Gamma^{DM}_{it} = f_t(x_i)$ and $\Gamma^{DR}_{it} = f_t(x_i) + \frac{\mathbb{1}\{T_i = t\}}{p_{it}}\big(y_i - f_t(x_i)\big)$, which stays consistent if either f_t or p_{it} is right [7].
4. *Policy tree.* Solve $\hat{\pi} = \arg\max_{\pi \in \Pi_D} \frac{1}{n} \sum_i \Gamma_{i,\pi(x_i)}$ over trees of depth $D \le 3$ [2].

Why this method. The setting is exactly contextual decisions with logged actions and a finite action set, and strategists need rules they can audit, which a shallow optimal tree gives and a black box does not.

Challenges. *Overlap:* DR divides by p_{it}, which is near zero where pitting almost never happens (very early laps), so rewards explode. We will plot $\hat{p}_{it}$ by action and lap phase, then restrict to common

support (reporting how many rows are dropped) or clip weights. *Unobserved confounding:* teams use private pace models, tyre temperatures and driver feedback we cannot see; if these drive both the decision and the outcome, no estimator is unbiased. We name this, discuss its likely direction (strong teams both pit well and have fast cars), and add team, driver and circuit effects. *Other:* rivals react to my stop (interference); tyre age and fuel burn move together; pitting under safety car is usually the optimal choice; mandatory use of two dry compounds of the tyres restricts the decisions (we include compounds used so far in x_i); regulation changes limit pooling across seasons.

3. Evaluation

We cannot re-run races, so we use off-policy evaluation. We split by *race* (grouped k-fold plus a hold-out season), never by lap, to avoid leakage within a race. On held-out races, with nuisance models trained on training races only, we estimate $\hat{V}(\pi) = \frac{1}{n_{\text{test}}} \sum_i \Gamma^{DR}_{i,\pi(x_i)}$ and report the gain over history in seconds vs. the field over K laps, with bootstrap intervals over races.

Baselines.

1. *Historical strategy:* the behaviour to beat; any gain is relative to what teams did.
2. *Threshold rule* (pit when tyre age exceeds a threshold tuned on training races): shows whether the tree beats the simplest strategist heuristic.
3. *DM tree vs. DR tree:* tests whether the propensity correction matters.
4. *Black-box policy* (gradient boosting on Γ): an upper reference quantifying what interpretability costs.

Robustness and sanity checks. We report estimated gain, tree depth, and results by circuit type and season, DM vs. DR, and sensitivity to K, to the clipping level and to safety-car rows. We also check that rules make racing sense and recover known facts (softer compounds degrade faster; stops under safety cars are cheaper). Implausible rules or very large gains will be treated as signs of overlap or confounding problems rather than discoveries.

References

[1] James Bekker and Willem Lotz. Planning Formula One race strategies using discrete-event simulation. *Journal of the Operational Research Society*, 60(7):952–961, 2009.

[2] Dimitris Bertsimas and Jack Dunn. Optimal classification trees. *Machine Learning*, 106(7):1039–1082, 2017.

[3] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

[4] Alexander Heilmeier, Michael Graf, Johannes Betz, and Markus Lienkamp. Application of monte carlo methods to consider probabilistic effects in a race simulation for circuit motorsport. *Applied Sciences*, 10(12):4229, 2020.

[5] Alexander Heilmeier, André Thomaser, Michael Graf, and Johannes Betz. Virtual strategy engineer: Using artificial neural networks for making race strategy decisions in circuit motorsport. *Applied Sciences*, 10(21):7805, 2020.

[6] OpenF1. OpenF1: Open-source Formula 1 data API. https://openf1.org; REST base URL https://api.openf1.org/v1, 2026. Accessed October 2026.

[7] James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89 (427):846–866, 1994.